TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	90.8 ± 0.87
Correlation Stability	94.1 ± 0.54
Distribution Consistency	92.0 ± 1.90
Entropy Preservation	95.4 ± 0.80
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.4 ± 0.66
Data Density	100.0 ± 0.00
Coverage Score	96.3 ± 1.42
Feature Accuracy	95.6 ± 1.20
Novelty Score	0.8 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	53.434	-1.005	1.8%	OK
sex	0.680	0.636	-0.044	6.4%	OK
cp	3.158	3.178	+0.019	0.6%	OK
trestbps	131.690	131.161	-0.529	0.4%	OK
chol	246.693	249.393	+2.699	1.1%	OK
fbs	0.149	0.132	-0.016	11.0%	Warn
restecg	0.990	0.996	+0.006	0.6%	OK
thalach	149.607	150.988	+1.380	0.9%	OK
exang	0.327	0.285	-0.042	12.7%	Warn
oldpeak	1.040	0.953	-0.086	8.3%	OK
slope	1.601	1.521	-0.080	5.0%	OK
ca	0.672	0.657	-0.015	2.3%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	90.8 ± 0.87
Correlation Stability (*2)	94.1 ± 0.54
Distribution Consistency (*3)	92.0 ± 1.90
Entropy Preservation (*4)	95.4 ± 0.80
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.4 ± 0.66

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	96.3 ± 1.42
Feature Accuracy (*10)	95.6 ± 1.20
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5311475409836065	0.5819672131147542	0.5754098360655738	0.0442622950819672	
F1	0.5180691030958732	0.5603350985186323	0.5594869957705086	0.04141789267463544	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.5934426229508196	0.562295081967213	0.01967213114754096	
F1	0.5198013329305267	0.5732798797034291	0.5552594802939025	0.03545814736337575	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5245901639344261	0.5131147540983607	0.5131147540983606	-0.011475409836065598	
F1	0.5306055374473726	0.521721792498271	0.5271314602226645	-0.00347407722470802	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.5606557377049181	0.5688524590163935	0.011475409836065587	
F1	0.568660979990255	0.5636013936621815	0.5776817383671766	0.009020758376921595	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5836065573770493	0.5721311475409836	-0.03606557377049182	
F1	0.6070236204175147	0.5806039968328489	0.5655418760101283	-0.04148174440738621	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5754098360655738	0.5967213114754099	0.6065573770491803	0.03114754098360656	
F1	0.5341186482274947	0.555394142234831	0.5679709804360196	0.03385233220852492	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5852459016393443	0.5901639344262295	-0.00819672131147549	
F1	0.5648347697855012	0.5566623041949946	0.563740571520291	-0.0010941982652100668	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5737704918032787	0.5590163934426229	0.6213114754098361	0.04754098360655736	
F1	0.5427152795354051	0.5399593734147267	0.5790816659907304	0.03636638645532536	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5163934426229508	0.518032786885246	-0.0360655737704918	
F1	0.4442202071436568 4	0.4236824032309817	0.4311881896952657	-0.013032017448391164	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	4.0	117.0	253.0	0.0	0.0	142.0	1.0	3.6
2	70.0	1.0	4.0	144.0	200.0	1.0	1.0	141.0	0.0	3.4
3	57.0	1.0	3.0	120.0	201.0	0.0	0.0	140.0	0.0	0.0
4	35.0	1.0	4.0	138.0	204.0	1.0	0.0	156.0	1.0	0.0
5	51.0	0.0	3.0	120.0	244.0	0.0	0.0	142.0	0.0	0.5
6	51.0	1.0	2.0	125.0	282.0	0.0	2.0	174.0	0.0	0.0
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	40.0	1.0	1.0	120.0	235.0	0.0	2.0	182.0	0.0	1.0
2	65.0	0.0	3.0	146.0	278.0	0.0	2.0	172.0	0.0	1.4
3	57.0	1.0	4.0	120.0	276.0	0.0	2.0	103.0	1.0	2.1
4	40.0	1.0	4.0	120.0	198.0	0.0	0.0	161.0	0.0	0.0
5	48.0	1.0	4.0	130.0	243.0	1.0	2.0	126.0	1.0	1.0
6	56.0	0.0	2.0	135.0	288.0	0.0	0.0	153.0	0.0	1.2
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	58.0	1.0	4.0	110.0	239.0	0.0	0.0	126.0	1.0	2.5
2	60.0	1.0	2.0	128.0	270.0	0.0	2.0	115.0	0.0	0.3
3	67.0	1.0	1.0	134.0	228.0	1.0	2.0	138.0	0.0	1.8
4	67.0	1.0	3.0	115.0	299.0	0.0	2.0	160.0	0.0	1.8
5	53.0	0.0	3.0	136.0	216.0	0.0	0.0	162.0	0.0	1.6
6	57.0	1.0	3.0	125.0	168.0	0.0	0.0	174.0	0.0	1.6
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	45.0	0.0	3.0	130.0	231.0	0.0	0.0	149.0	0.0	0.3
2	49.0	1.0	4.0	130.0	261.0	0.0	2.0	188.0	0.0	1.2
3	45.0	1.0	4.0	132.0	298.0	1.0	0.0	150.0	1.0	0.1
4	58.0	1.0	4.0	128.0	330.0	1.0	1.0	140.0	1.0	1.2
5	44.0	1.0	3.0	138.0	315.0	0.0	0.0	171.0	0.0	0.0
6	41.0	1.0	3.0	135.0	175.0	0.0	0.0	126.0	0.0	0.0
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	61.0	0.0	4.0	145.0	241.0	0.0	0.0	138.0	1.0	1.4
2	60.0	1.0	4.0	117.0	300.0	0.0	2.0	142.0	1.0	0.0
3	53.0	0.0	4.0	138.0	264.0	0.0	0.0	160.0	1.0	0.0
4	62.0	1.0	4.0	135.0	261.0	0.0	0.0	169.0	0.0	0.0
5	70.0	1.0	4.0	110.0	237.0	0.0	2.0	136.0	1.0	2.8
6	51.0	1.0	4.0	128.0	290.0	0.0	2.0	171.0	0.0	0.0
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	48.0	1.0	4.0	130.0	212.0	0.0	2.0	168.0	0.0	0.9
2	67.0	1.0	3.0	135.0	234.0	0.0	0.0	155.0	0.0	0.0
3	59.0	1.0	4.0	140.0	232.0	0.0	2.0	127.0	1.0	1.2
4	59.0	1.0	3.0	154.0	274.0	0.0	2.0	160.0	0.0	2.8
5	63.0	1.0	2.0	146.0	211.0	1.0	0.0	159.0	0.0	0.2
6	53.0	1.0	4.0	123.0	188.0	1.0	2.0	146.0	1.0	0.0
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	47.0	1.0	2.0	101.0	275.0	0.0	2.0	152.0	0.0	1.0
2	53.0	1.0	4.0	138.0	226.0	0.0	2.0	143.0	1.0	3.0
3	50.0	0.0	3.0	120.0	219.0	0.0	0.0	185.0	0.0	1.6
4	42.0	0.0	2.0	120.0	341.0	0.0	0.0	179.0	0.0	1.2
5	65.0	1.0	3.0	146.0	282.0	1.0	2.0	158.0	0.0	1.4
6	64.0	0.0	3.0	140.0	303.0	0.0	0.0	157.0	0.0	1.4
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	1.0	3.0	120.0	197.0	0.0	0.0	105.0	0.0	0.0
2	61.0	0.0	4.0	130.0	236.0	0.0	0.0	159.0	0.0	1.2
3	54.0	1.0	3.0	108.0	240.0	1.0	2.0	148.0	0.0	0.4
4	57.0	0.0	4.0	140.0	305.0	0.0	0.0	174.0	1.0	0.0
5	44.0	0.0	3.0	134.0	308.0	0.0	2.0	178.0	0.0	0.0
6	61.0	0.0	4.0	130.0	307.0	0.0	2.0	155.0	1.0	0.6
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	66.0	0.0	3.0	155.0	360.0	0.0	2.0	172.0	0.0	0.0
2	52.0	1.0	3.0	130.0	255.0	1.0	0.0	166.0	0.0	0.4
3	41.0	1.0	4.0	110.0	199.0	0.0	2.0	168.0	0.0	0.0
4	66.0	0.0	4.0	155.0	360.0	0.0	2.0	159.0	0.0	0.0
5	60.0	1.0	4.0	174.0	231.0	1.0	2.0	182.0	0.0	0.0
6	54.0	1.0	2.0	134.0	232.0	0.0	2.0	165.0	0.0	0.6
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	65.0	1.0	4.0	120.0	211.0	0.0	0.0	154.0	0.0	0.4
2	41.0	1.0	4.0	110.0	198.0	0.0	0.0	130.0	1.0	1.9
3	54.0	1.0	4.0	122.0	212.0	0.0	2.0	108.0	1.0	0.0
4	57.0	1.0	4.0	100.0	126.0	0.0	0.0	147.0	0.0	0.1
5	66.0	1.0	1.0	160.0	212.0	0.0	2.0	115.0	1.0	1.8
6	45.0	0.0	2.0	130.0	236.0	0.0	2.0	184.0	1.0	0.4
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6